

AASAIMANI THAMIZHAZHAGAN

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ACADEMIC POSITION(S)

- Pondicherry University** • Kalapet, Pondicherry, India Oct 2024 – April 2026
Guest Faculty • Department of Mathematics
- University of Waterloo** • Waterloo, ON, Canada Sept 2022 – May 2024
Postdoctoral Fellow and Sessional Instructor • Department of Pure Mathematics
Worked with: [Dr. Nico Spronk](#), [Dr. Brian E. Forrest](#)
- Conestoga College** • Kitchener, ON, Canada Aug 2023 – Jan 2024
Part-time Faculty • School of Business
- University of Winnipeg** • Winnipeg, MB, Canada Sept 2021 – Aug 2022
Postdoctoral Fellow and Contract Academic Staff • Department of Mathematics and Statistics
Worked with : [Dr. Ross Stokke](#), [Dr. Matthew Wiersma](#)
Funding: Joint support from grants held by the host researchers, [Dr. Yong Zhang](#), and [Dr. Fereidoun Ghahramani](#)

EDUCATION

- University of Waterloo** • Waterloo, ON, Canada Sep 2016 – Aug 2021
Doctor of Philosophy • Pure Mathematics
Thesis - [On the structure of invertible elements in certain Fourier-Stieltjes algebras](#)
Advisors: [Dr. Nico Spronk](#), [Dr. Brian E. Forrest](#)
- National Institute of Science Education and Research (NISER)** • Odisha, India July 2011 – May 2016
Integrated Masters of Science • Mathematics
Thesis - Fourier algebras, amenability and its relations with representation theory
Advisor: Dr. [Dr. Varadharajan Muruganandam](#)

TEACHING AND RESEARCH INTERESTS

Teaching: Mathematical proofs, single- and multi-variable calculus, linear algebra, real analysis, complex analysis, functional analysis, measure theory and Lebesgue integration, harmonic analysis, topology, and group theory.

Research: *Functional analysis and its intersection with group theory.*
Abstract harmonic analysis, representation theory of locally compact groups, Banach algebras, operator algebras and operator space theory.

PUBLICATIONS

- (with Brian E. Forrest and John Sawatzky) *Arens regularity of ideals in $A(G)$, $A_{cb}(G)$ and $A_M(G)$*
Journal of the Iranian Mathematical Society 4 (2023), no. 1, 5–25.   
- (with Brian E. Forrest and John Sawatzky) *Invariant subspaces in the dual of $A_{cb}(G)$ and $A_M(G)$*
Annals of Mathematical Sciences and Applications 8 (2023), no.2, 239–267.   
- On the structure of invertible elements in certain Fourier-Stieltjes algebras
Studia Mathematica, 257 (2021), no. 3, 347–360.   

INVITED RESEARCH TALKS:

- Homomorphisms of Spine Algebras**
Mathematics Seminar, Institute of Mathematical Sciences, Chennai Jan 9, 2026
Workshop on Commutative and Non-commutative Harmonic Analysis, NISER, Bhubaneswar Mar 7, 2026
- On the structure of invertible elements in certain Fourier-Stieltjes Algebra**
Canadian Abstract Harmonic Analysis Symposium, Banff International Research Station June 19, 2022
Department Seminar Series, University of Winnipeg Sep 24, 2021
Groups, Operators, and Banach Algebras Webinar, Online Aug 18, 2020
Analysis Seminar in Pure Mathematics, University of Waterloo Sep 18, 2019

INVITED COLLOQUIA:

- **On the interplay of harmonic analysis, combinatorics, additive number theory, and ergodic theory**
Joint PM/CO Grad Colloquium, University of Waterloo Dec 3, 2020
- **Uncertainty principles and Fourier analysis**
Grad Student Colloquium, University of Waterloo Nov 27, 2018

CONTRIBUTED TALKS:

- **Homomorphisms of Spine Algebras**
19th Discussion Meeting in Harmonic Analysis, IISER Mohali Dec 16, 2025
- **On the structure of invertible elements in certain Fourier-Stieltjes Algebra**
National Seminar on Algebra and Graph Theory, Pondicherry University Feb 22, 2025
- **Invariant subspaces in the Dual of $A_{cb}(G)$ and $A_M(G)$**
CAHAS 2023, University of St. Boniface June 15, 2023
- **On the structure of invertible elements in certain Fourier-Stieltjes Algebra**
Virtual Math Fest, Institute of Mathematical Sciences, Chennai (Online) July 24, 2020
Banach Algebras and Applications, University of Manitoba July 18, 2019
Grad Seminar in Pure Mathematics, University of Waterloo May 15, 2019

PRESENTATIONS:

- **Analytic discs in the maximal ideal space of H^∞**
Grad course talks on Hardy spaces, University of Waterloo Dec, 2017
- **Perturbations of approximately finite-dimensional C^* -algebras**
Grad course talks on K -theory of C^ -algebras*, University of Waterloo April, 2017

CONFERENCES/WORKSHOPS/SEMINARS PARTICIPATION

- [Workshop on Commutative and Non-commutative Harmonic Analysis](#), NISER Bhubaneswar Mar 2–7, 2026
- [National Seminar on Algebra and Graph Theory](#), Pondicherry University, Puducherry Feb 18–21, 2026
- [19th Discussion Meeting in Harmonic Analysis](#), IISER Mohali Dec 16–19, 2025
- [International Conference on Special Functions & Applications \(ICSFA-2025\)](#)
Pondicherry University, Puducherry Nov 20–22, 2025
- [National Workshop on Python for Scientific Computing and Applications \(PSCA-2025\)](#)
Maulana Azad National Institute of Technology (MANIT), Bhopal (Online) June 16–20, 2025
- [National Seminar on Algebra and Graph Theory](#), Pondicherry University, Puducherry Feb 19–22, 2025
- [Canadian Abstract Harmonic Analysis Symposium\(CAHAS\)](#),
University of St. Boniface, Winnipeg June 15–16, 2023
- [A Celebration of Great Math Education Initiatives](#), FYMSiC Online Meet Up Feb 28, 2023
- [Canadian Abstract Harmonic Analysis Symposium\(CAHAS\) 2020](#), BIRS, Banff June 17–19, 2022
- [Virtual Math Fest](#), Institute of Mathematical Sciences, Chennai (Online) July 20–26, 2020
- [Canadian Operator Symposium \(COSy\)](#), Fields Institute, Online May 25–29, 2020
- [Banach Algebras and Applications](#), University of Manitoba, Winnipeg July 11–18, 2019
- [Canadian Abstract Harmonic Analysis Symposium\(CAHAS\)](#),
Carleton University, Ottawa May 31–June 2, 2018
- [Canadian Abstract Harmonic Analysis Symposium\(CAHAS\)](#),
University of Manitoba, Winnipeg May 23–25, 2017
- [Canadian Operator Symposium\(COSy\)](#), Lakehead University, Thunder Bay May 29–June 2, 2017
- [Workshop on Fourier and signal analysis by G. B. Folland](#), NISER, Bhubaneswar Dec 2015

TEACHING EXPERIENCE

Pondicherry University, Ramanujam School of Mathematical Sciences

Instructor/Sole Instructor

- *Mathematical Modeling (BSMT 331)* (Sole Instructor) - Int. B.Sc. (Hons) Even 25-26
- *Mathematics I for Engineers (BTMT 171)* Odd 2024-25, 25-26
- *Mathematics II for Engineers (BTMT 172)* Even 2024-25, 25-26

ECE, CSE, ENE; $\sim$ 120 students

Topics: linear algebra, ODEs, multivariable calculus, PDEs, Laplace and Fourier transforms. Coordinated syllabi, assessments, and examinations across multiple engineering cohorts.

University of Waterloo, Faculty of Mathematics

Instructor/Sole Instructor

- *Advanced Calculus II for Electrical Engineers (ECE 206/MATH 212)* (Sole Instructor) Fall 2023
Multivariable and vector calculus; complex variables (Möbius-based curriculum).
- *Calculus I for Honours Mathematics (MATH 137)* Fall 2022
- *Calculus for Engineering (MATH 116)* Fall 2019

University of Winnipeg, Department of Mathematics & Statistics

Instructor/Sole Instructor

- *Complex Analysis (MATH 4101)* (Sole Instructor) Winter 2022
- *Applied Mathematics for Business & Administration (MATH 1301)* Fall 2021

Conestoga College, School of Business

Instructor

- *Business Mathematics (MATH 71775)*

University of Waterloo, Department of Pure Mathematics

Graduate Teaching Assistant

2016-2021

- Tutorials and grading for calculus and linear algebra
- Academic support for upper-year analysis courses: real analysis, measure theory, Fourier analysis, complex analysis.

PROFESSIONAL/ACADEMIC DEVELOPMENT

CSIR-UGC NET Coach for Math Majors – Professional Service

May 2025 – April 2026

Pondicherry University, Equal Opportunity Cell

- Provided personalised coaching to undergraduate and postgraduate students, focusing on conceptual clarity, problem-solving strategies, and exam readiness for CSIR-UGC NET-mathematics subject.

Teaching and Curriculum Development Training –Micro-credentials

Summer 2023

Conestoga College, Teaching and Learning Department

- Completed certified micro-credentials in outcomes-based education, active learning, assessment design, and curriculum alignment.

Graduate Student Instructor Training Seminar

Winter 2019

University of Waterloo, Faculty of Mathematics

- Participated in interactive teaching workshops with peer feedback on lesson plans and course materials.
- Attended math-specific sessions on lesson planning, assessment design, student motivation, and lecture delivery.
- Designed and presented three mini-lectures in real analysis and observed multiple classroom teaching sessions.

Learning Seminars

Fall 2016 – Summer 2024

University of Waterloo, Department of Pure Mathematics

- Contributed to and participated in research seminars in operator algebras, harmonic analysis and representation theory, including Banach algebras associated with locally compact groups, C^* -algebras, and tensor categories.
- Regular attendee departmental analysis seminars, graduate colloquia, and graduate seminars.
- Supervision of undergraduate research assistants

AWARDS AND HONOURS

Received at University of Waterloo, Waterloo, ON, Canada

- International Doctoral Student Award Fall 2016 – Spring 2020
- Susan and Janoes Aczel Graduate Scholarship Winter 2018 – 2020 and Fall 2018

Received at National Institute of Science Education and Research, Odisha, India

- INSPIRE Fellowship July 2011 – May 2016
- Summer Research Fellowship, Indian Academy of Sciences, Bangalore, India Summer 2013
- Cleared CSIR-UGC-NET for JRF with rank 33 June 2016